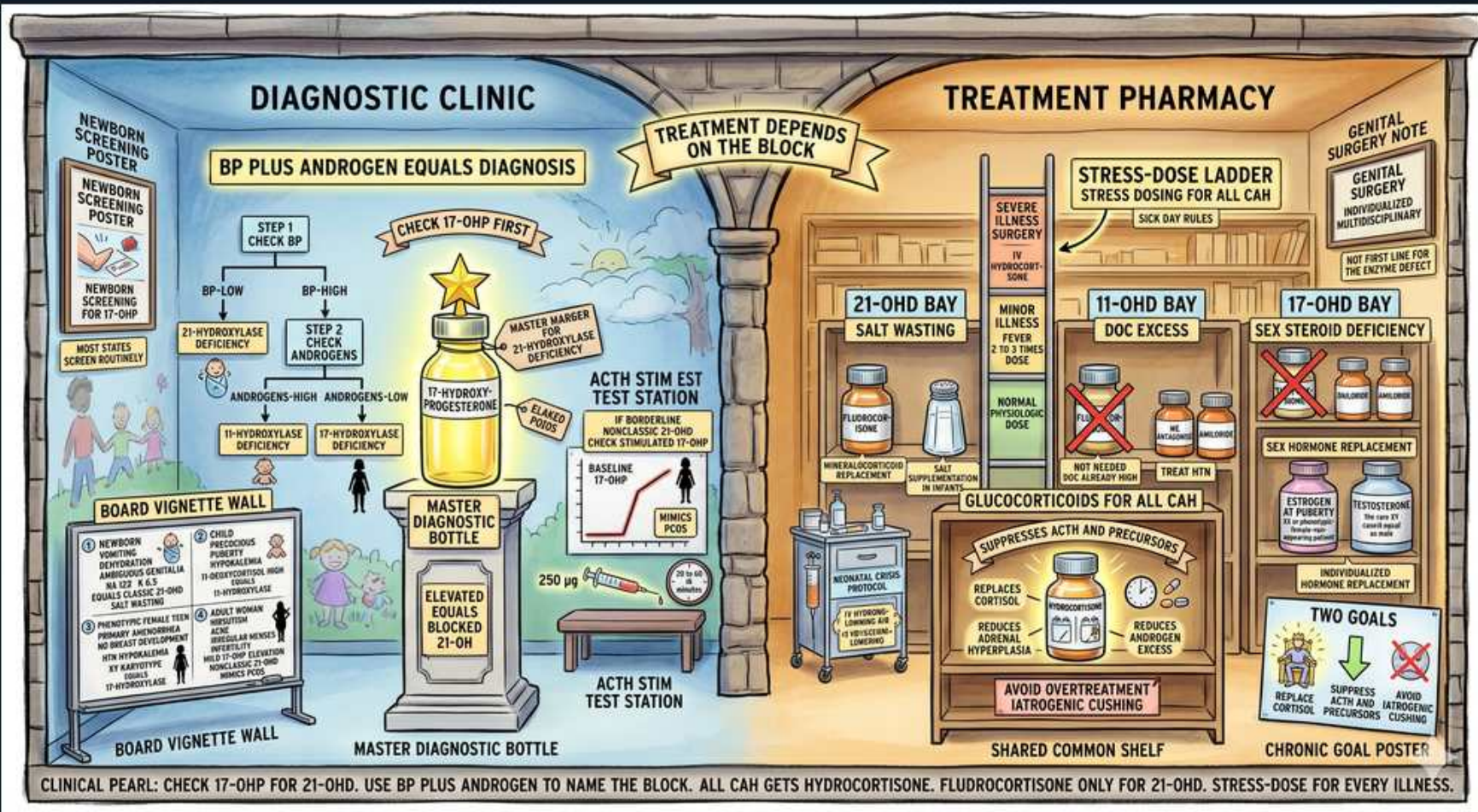






## 1. BP + androgen = diagnosis

### WHAT YOU SEE

Banner: BP plus androgen equals diagnosis

### WHAT IT ENCODES

The whole CAH workup collapses to two axes: BLOOD PRESSURE (low = no mineralocorticoid = 21-OH; high = DOC excess = 11- or 17-OH) and ANDROGENS (high = 21- or 11-OH; low = 17-OH). Cross-referencing the two uniquely names the blocked enzyme.

### BOARD TRAP

Don't memorize each enzyme in isolation — solve it as a 2x2 of BP and androgens and the answer falls out.

## 2. Newborn screening

### WHAT YOU SEE

Newborn screening poster on the left wall

### WHAT IT ENCODES

Classic 21-hydroxylase deficiency is detected on universal NEWBORN SCREENING via a heel-stick 17-OHP level, allowing treatment before a salt-wasting crisis. A confirmatory cosyntropin-stimulated 17-OHP and electrolytes follow a positive screen.

### BOARD TRAP

Screening measures 17-OHP — it catches 21-OH CAH; it does not screen for the rarer 11- or 17-OH forms.

## 3. Step 1: check BP

### WHAT YOU SEE

Algorithm Step 1 CHECK BP branching BP-low vs BP-high

### WHAT IT ENCODES

Step 1 — BP/volume status splits the tree: LOW BP with hyponatremia/hyperkalemia (salt-wasting) → 21-hydroxylase deficiency; HIGH BP with hypokalemia (DOC-driven) → 11- or 17-hydroxylase deficiency.

### BOARD TRAP

Hypertension at presentation immediately excludes 21-OH and points to an 11- or 17-OH block.

## 4. Step 2: check androgens

### WHAT YOU SEE

Step 2 CHECK ANDROGENS, androgens-high vs androgens-low

### WHAT IT ENCODES

Step 2 — among the high-BP forms, androgens separate them: HIGH androgens/virilization = 11-hydroxylase; LOW androgens/absent puberty = 17-hydroxylase. (Among low-BP forms, 21-OH has high androgens.)

### BOARD TRAP

High BP + virilization = 11-OH; high BP + NO sexual development = 17-OH — androgens are the tiebreaker.

## 5. Check 17-OHP first

### WHAT YOU SEE

Note CHECK 17-OHP FIRST near the master bottle

### WHAT IT ENCODES

Because 21-OH deficiency is ~90–95% of CAH, the FIRST and master test is a 17-OHP level; a markedly elevated baseline 17-OHP is essentially diagnostic of classic 21-hydroxylase deficiency.

### BOARD TRAP

17-OHP is specific to the 21-OH block — it is normal in 11- and 17-OH deficiency, so a normal 17-OHP does not exclude CAH if BP is high.

## 6. Elevated 17-OHP bottle

### WHAT YOU SEE

Glowing master diagnostic bottle: elevated equals blocked 21-OH

### WHAT IT ENCODES

A high BASELINE 17-OHP (typically well above the screening cutoff) confirms classic 21-OH CAH without further testing; the precursor backs up directly above the enzyme block.

## 7. ACTH stim test station

### WHAT YOU SEE

ACTH STIM TEST station with curve, used when borderline

### WHAT IT ENCODES

When baseline 17-OHP is borderline (typically the nonclassic/late-onset PCOS-mimic), a cosyntropin (synthetic ACTH) STIMULATION test is done: an exaggerated rise in 17-OHP after ACTH confirms the diagnosis.

### BOARD TRAP

Use the ACTH stim test to confirm NONCLASSIC 21-OH CAH when baseline 17-OHP is equivocal — a hirsute young woman with borderline 17-OHP.

## 8. Glucocorticoids for ALL CAH

### WHAT YOU SEE

Shared common shelf: glucocorticoids for all CAH

### WHAT IT ENCODES

Every CAH gets a GLUCOCORTICOID — it does double duty: replaces the deficient cortisol AND, by restoring negative feedback, lowers ACTH so the adrenal stops overproducing whatever precursor/androgen/DOC is causing harm. Hydrocortisone is preferred in children, longer-acting agents in adults.

### BOARD TRAP

Glucocorticoid is the common backbone of ALL CAH treatment regardless of which enzyme is missing.



## 9. 21-OH salt-wasting shelf

### WHAT YOU SEE

21-OH BAY shelf labeled salt wasting

### WHAT IT ENCODES

Salt-wasting 21-OH CAH lacks aldosterone, so on top of glucocorticoid it needs a MINERALOCORTICOID — FLUDROCORTISONE — plus sodium chloride supplementation in infancy. This restores Na<sup>+</sup> retention and corrects the hyperkalemia/hypotension.

### BOARD TRAP

ONLY the salt-wasting (21-OH) form needs fludrocortisone; 11- and 17-OH have mineralocorticoid (DOC) EXCESS, so adding fludrocortisone would worsen their HTN.

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## 10. 11-OH DOC-excess shelf

### WHAT YOU SEE

11-OH BAY labeled DOC excess

### WHAT IT ENCODES

In 11-OH (and 17-OH) deficiency the HTN comes from DOC excess; giving a glucocorticoid suppresses ACTH → less DOC → blood pressure and potassium normalize. No fludrocortisone is given (they already retain salt).

### BOARD TRAP

For the high-BP forms, the glucocorticoid itself fixes the hypertension by lowering ACTH/DOC — adding a mineralocorticoid is wrong.

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## 11. Stress-dose ladder

### WHAT YOU SEE

STRESS-DOSE LADDER, stress dosing for all CAH, sick-day rules

### WHAT IT ENCODES

Because their adrenals cannot mount a cortisol surge, ALL CAH patients need STRESS-DOSE steroids (sick-day rules): roughly double/triple oral hydrocortisone for febrile illness, and IV hydrocortisone for surgery, trauma or vomiting, to prevent an adrenal crisis.

### BOARD TRAP

Failing to stress-dose during illness/surgery — or the patient stopping steroids — precipitates adrenal crisis; never skip it.

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## 12. Avoid overtreatment

### WHAT YOU SEE

Warning: avoid overtreatment iatrogenic Cushing, suppressed ACTH and precursors

### WHAT IT ENCODES

Therapy is a balance: OVER-replacement (precursors/17-OHP fully suppressed) causes iatrogenic Cushing, weight gain and growth suppression; UNDER-replacement lets ACTH-driven androgens/precursors rise → virilization and accelerated bone age. Monitor 17-OHP and androgens, growth and bone age.

### BOARD TRAP

Completely normalizing 17-OHP signals OVER-treatment — aim for mildly elevated, not suppressed.

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## 13. Two goals poster

### WHAT YOU SEE

Chronic goal poster: two goals — replace cortisol AND suppress androgens

### WHAT IT ENCODES

The twin chronic goals: (1) REPLACE the deficient cortisol ( $\pm$  aldosterone in salt-wasters) and (2) SUPPRESS the ACTH-driven excess androgens/precursors to a near-normal range — without tipping into iatrogenic Cushing.

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